

1 P1 C4: five lines in $\mathbb{R}^3$ and six in $\mathbb{R}^4$

For N lines through the origin, choose unit representatives $x_1, \dots, x_N$, permitting repeated lines. Their acute pairwise angles and total angle sum are

$$\theta(x, y) = \arccos |x \cdot y| \in [0, \pi/2], \quad S = \sum_{i < j} \theta(x_i, x_j).$$

We prove the sharp bounds $S \leq 4\pi$ for five lines in $\mathbb{R}^3$ and $S \leq 13\pi/2$ for six lines in $\mathbb{R}^4$. Put

$$\delta(x, y) = \arcsin |x \cdot y| = \frac{\pi}{2} - \theta(x, y), \quad D = \sum_{i < j} \delta(x_i, x_j).$$

In both cases the requested bound is equivalent to $D \geq \pi$. We prove that inequality for every configuration and exhibit equality in each dimension. This is a written mathematical argument; selected supporting lemmas are formalized in Lean, but the complete C4 proof is not.

1.1 1. A maximizing configuration can be made sparse

For fixed N, d , the angle sum attains a maximum on the compact product of N unit spheres. Among maximizers choose one with the largest number of orthogonal pairs.

For a vector x with at least one nonorthogonal neighbor, let W span its orthogonal neighbors. If $\dim W < d - 1$, choose a unit y orthogonal to W and to x . Move x along $x(t) = x \cos t + y \sin t$. Every existing orthogonality is preserved. For any other vector z not orthogonal to x , its inner product is $A \cos t + B \sin t$, with $A \neq 0$. On the interval between its consecutive zeros containing $t = 0$, its line angle is convex. To check this, write the signed inner product as $r \cos(t - t_0) > 0$, $0 < r \leq 1$. For $r < 1$,

$$\frac{d^2}{dt^2} \arccos(r \cos t) = \frac{r(1 - r^2) \cos t}{(1 - r^2 \cos^2 t)^{3/2}} \geq 0.$$

For $r = 1$ the angle is an absolute-value function on that interval, also convex. Intersect these zero-free intervals for all nonorthogonal neighbors. The total angle sum is convex there, and has its global maximum at the interior point 0, so it is constant. At an endpoint at least one additional pair becomes orthogonal, while no orthogonal pair is lost. This contradicts the choice of maximizer.

Here $N > d$, so at least one pair is nonorthogonal. If the configuration spanned dimension $r < d$, an endpoint x of such a pair would have $\dim W \leq r - 1 < d - 1$, contradicting the preceding rotation argument. Thus the configuration spans $\mathbb{R}^d$. For a vector orthogonal to every other vector, those other vectors consequently span its entire orthogonal complement; its orthogonal-neighbor span has dimension $d - 1$ too. Every vector therefore has at least $d - 1$ linearly independent orthogonal neighbors. Its nonorthogonality graph (an edge means nonzero inner product) has maximum degree at most $N - d$.

For $N = d + 1$, this graph consists of edges and isolated vertices. The Gram matrix has nullity one. Some two-vertex block must have rank one, hence its two lines coincide and contribute deficiency $\pi/2$. We have proved the auxiliary bound:

For $m \geq 2$, any m unit vectors spanning dimension at most $m - 1$ have total deficiency at least $\pi/2$. This is the auxiliary bound (A).

To apply it in smaller dimension, embed that space in $\mathbb{R}^{m-1}$.

For $N = d+2$, the sparse graph consists of paths and cycles. Gram blocks belonging to different components are orthogonal. An irreducible path block has nullity at most one: the first row and the tridiagonal recurrence determine every kernel coordinate from the first coordinate. An irreducible cycle block has nullity at most two: two adjacent coordinates determine all others. The whole matrix has nullity exactly two. If two components are singular, (A) gives deficiency at least $\pi/2$ in each, proving the target. Otherwise one cycle has nullity two, and all other components are nonsingular.

For $N \leq 6$ only cycles of lengths 3, 4, 5, 6 remain.

A three-cycle of nullity two has rank one: all three lines coincide and its deficiency is $3\pi/2$. A four-cycle of nullity two has rank two: its opposite pairs are orthonormal bases of the same plane. For every unit planar vector, its acute angles to an orthogonal pair sum to $\pi/2$. Consequently the four edge deficiencies sum to π .

1.2 2. Two elementary analytic inequalities

For $p, q \in [0, \pi/2]$, put

$$h = \arccos(\cos p \cos q),$$

$$E(p, q) = \arcsin\left(\frac{\sin p \sin q}{1 + \cos p \cos q}\right).$$

Then

$$h + E(p, q) \leq p + q. \tag{B}$$

Here is a direct proof, with no geometric theorem required. Set $A = \cos p$, $B = \cos q$, $r = \sin p \sin q$, $u = AB$, and $L = \sqrt{1 - u^2}$. The identity

$$(1 + u)^2 - r^2 = (A + B)^2.$$

gives $\cos E = (A + B)/(1 + u)$. Therefore

$$(1 + u)[\cos(h + E) - \cos(p + q)] = -u(1 - A)(1 - B) + r(1 + u - L) \geq 0.$$

Indeed $L \leq 1$ and $r \geq (1 - A)(1 - B)$, since $\sin t \geq 1 - \cos t$ on $[0, \pi/2]$. Both arguments compared lie in $[0, \pi]$, where cosine is decreasing. This proves (B). The Lean declaration `C4Pentagon.pentagon_scalar_angles` proves this inequality.

A second inequality, for $T, p, q \in [0, \pi/2]$, is

$$\arcsin(\sin p \sin q) \leq \arccos(\cos T \cos p) + \arccos(\sin T \cos q) - \frac{\pi}{2}. \tag{C}$$

Set $X = \cos T \cos p$, $Y = \sin T \cos q$, and $Z = \sin p \sin q$. Direct expansion gives

$$X^2 + Y^2 + Z^2 + 2XYZ - 1 = -(\cos T \sin p \cos q - \sin T \sin q \cos p)^2 \leq 0.$$

For nonnegative $X, Y, Z \leq 1$ this implies

$$\sqrt{1-X^2}\sqrt{1-Y^2} \geq Z + XY.$$

hence $\cos(\arccos X + \arccos Y) \leq -Z$. Since $\arccos X + \arccos Y \in [0, \pi]$, cosine monotonicity gives their sum at least $\pi - \arccos Z$. This is (C).

1.3 3. The five-cycle

Its Gram rank is three. Take $x_1 = e_1, x_3 = e_2$, and choose signs of coordinates so that, in absolute value,

$$x_4 = (0, c, \sqrt{1-c^2}), \quad x_5 = (a, 0, \sqrt{1-a^2}).$$

where $0 < a, c < 1$. Possible remaining signs do not affect any formula below. Since x_2 is orthogonal to x_4 and x_5 , their cross product determines it up to sign. With $L = \sqrt{a^2 + c^2 - a^2c^2}$, the absolute edge inner products are

$$\frac{c\sqrt{1-a^2}}{L}, \quad \frac{a\sqrt{1-c^2}}{L}, \quad c, \quad \sqrt{1-a^2}\sqrt{1-c^2}, \quad a.$$

Write $a = \sin p, c = \sin q$. If the first two deficiencies are U, V , then

$$\begin{aligned} \cos(U + V) &= \frac{\sin p \sin q}{1 + \cos p \cos q}, \\ \sin(U + V) &= \frac{\cos p + \cos q}{1 + \cos p \cos q}. \end{aligned}$$

Thus $U + V = \pi/2 - E(p, q)$. The middle product has deficiency $\pi/2 - \arccos(\cos p \cos q)$. The total edge deficiency is therefore

$$\pi + p + q - \arccos(\cos p \cos q) - E(p, q) \geq \pi.$$

by (B). The expressions extend continuously to boundary cases; alternatively those cases break the cycle and were already handled. This completes the five-line $\mathbb{R}^3$ bound.

1.4 4. Removing a vertex from the six-cycle

The six-cycle has Gram rank four. Remove x_6 and project x_1 and x_5 orthogonally onto $x_6^\perp$, normalizing the results. The other three vectors already lie in that hyperplane. The resulting five lines lie in $\mathbb{R}^3$.

Only four original deficiencies and three projected deficiencies need comparison. Relabel the local path $(x_2, x_1, x_6, x_5, x_4)$ as $(v_1, v_2, v_3, v_4, v_5)$, and choose signs making its consecutive inner products a, b, c, d positive. The odd vectors v_1, v_3, v_5 are orthonormal. In an orthonormal basis extending them,

$$v_2 = (a, b, 0, u), \quad v_4 = (0, c, d, v), \quad uv = -bc.$$

All $a, b, c, d, |u|, |v|$ are positive for an irreducible six-cycle. Choose the fourth basis direction so $u > 0$ and $v < 0$. Consequently there are $P, Q, T \in (0, \pi/2)$ with

$$\begin{aligned} a &= \cos P, & b &= \sin P \cos T, & u &= \sin P \sin T, \\ d &= \cos Q, & c &= \sin Q \sin T, & v &= -\sin Q \cos T. \end{aligned}$$

Let $B = \arcsin b, C = \arcsin c$. Let p, q be the acute angles of the normalized projections of v_2, v_4 to v_1, v_5 respectively. Then

$$\begin{aligned} \cos P &= \cos B \cos p, & \cos Q &= \cos C \cos q, \\ \tan B &= \sin p \cot T, & \tan C &= \sin q \tan T. \end{aligned}$$

The new connecting inner product has absolute value $\sin p \sin q$. Hence the original local deficiency is $\pi - P - Q + B + C$, whereas the projected local deficiency is $\pi - p - q + \arcsin(\sin p \sin q)$.

By (B),

$$B + p - P \geq E(B, p), \quad C + q - Q \geq E(C, q).$$

The tangent relations imply the following exact identities:

$$\begin{aligned} E(B, p) &= \arccos(\cos T \cos p) - T, \\ E(C, q) &= \arccos(\sin T \cos q) - \left(\frac{\pi}{2} - T\right). \end{aligned}$$

For completeness the first identity follows by substituting $\sin B \sin T = \cos B \sin p \cos T$ into the cosine addition formula for $E(B, p) + T$; the result is $\cos T \cos p$. Both angles are in $[0, \pi]$, making cosine injective. The second is the same computation with T replaced by $\pi/2 - T$.

Inequality (C) now gives

$$(B + p - P) + (C + q - Q) \geq \arcsin(\sin p \sin q).$$

Thus projection does not increase total deficiency. The projected five-line configuration has deficiency at least π by the result just proved. The original six-cycle therefore also has deficiency at least π .

Combining all component cases proves $D \geq \pi$ for the sparse angle maximizer. Every other configuration has no larger angle sum, so the same bound holds for every five- and six-line configuration. Equality is attained by coordinate axes with two axes repeated: multiplicities $(2, 2, 1)$ in $\mathbb{R}^3$ and $(2, 2, 1, 1)$ in $\mathbb{R}^4$.